Om Verma

Embedded Firmware Engineer

RESUME_1155

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Total Experience: 6 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 6 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: STM32, C, C++, Firmware Debugging, ARM Cortex-M, UART, Embedded Linux, I2C, CAN, Oscilloscope, SPI, FreeRTOS

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | SparrowCloud Labs | Jan 2023 - Present

- Reduced embedded task jitter by 26% using interrupt priority and queue tuning
- Improved battery runtime by 16% with peripheral and sleep-state optimization
- Lowered field-reset incidents by 40% with watchdog recovery and diagnostics
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | OrionPath Technologies | Jul 2020 - Dec 2022

- Lowered field-reset incidents by 24% with watchdog recovery and diagnostics
- Validated 822 hardware test vectors across protocol and timing corner cases

PROJECTS

Motor Control Interface | C, FreeRTOS, I2C, ARM Cortex-M

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Low-Power Metering Device | ARM Cortex-M, STM32, CAN, UART

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Northbridge College of Technology, 2020
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2022)